

# Motor Drive and Power Management Solutions

## Compelling Automotive IC Solutions that Put You in Control

### Brushless DC Motor Drivers (BLDC)

Allegro three phase controllers set the standard in the industry for high performance and robust features

### Brush DC Motor Drivers

Allegro full-bridge controllers combine to provide an arrangement of feature rich and versatile control for brush motor or other actuators that require full-bridge control using N-Channel MOSFETs

### Stepper Motor Drivers

Allegro bipolar stepper motor driver with integrated 1.5 A MOSFET outputs features a simple step and direction interface and excellent diagnostic and protection

### Power Management

Allegro offers a comprehensive product lineup to address the need for wide input and multiple output voltage specifications, high switching frequencies to avoid the AM band, and robust fault protection



|                            | Part Number   | Features                                                                                                                                                                                         |
|----------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brushless DC Motor Drivers | A3930 / A3931 | Three-phase N-Channel pre-driver with Hall inputs and 6 step commutation state sequencer                                                                                                         |
|                            | A4933         | Three-phase N-Channel pre-driver with charge pump, serial diagnostic read back capability, enhanced gate drive capability                                                                        |
|                            | A4935         | Three-phase N-Channel pre-driver with charge pump and serial diagnostic read back capability                                                                                                     |
| Brush DC Motor Drivers     | A3941         | Full-bridge N-Channel pre-driver with 5.5V to 50V input range and ENABLE and PHASE inputs                                                                                                        |
|                            | A4940         | Full-bridge N-Channel pre-driver with 5.5V to 50V input range and four direct drive inputs                                                                                                       |
| Stepper Motor Drivers      | A3980         | Integrated 1.5A MOSFET with step & direction interface                                                                                                                                           |
|                            | A3981         | Serial controlled stepper motor driver IC includes stall detect features and comprehensive diagnostics                                                                                           |
|                            | A4980         | Serial controlled stepper motor driver IC with low-voltage operation feature allowing the IC to run down to a load supply voltage of 3.3 V                                                       |
| Regulators                 | A8582/A8583   | Maximum Vin input voltage: 40 V supports load dump; Adjustable switching frequency: 250 KHz to 2.4 MHz; Power OK output; Sleep mode; Over voltage protection; Short to gnd tolerant at every pin |
|                            | A4402         | Integrated Power-On Reset and Watchdog Timer, adjustable soft-start, fixed on-time with up to 2 MHz operation, Overcurrent protection, 50 V supply voltage range                                 |
|                            | A8450         | Internal buck regulator with two linear regulators (5V tracking) and two linear controllers (3.3 V and 1.8 V to 3.3 V adjustable), Fault flag, 45V input supply range                            |

### Typical Applications include:

- Electronic Power Steering (EPS)
- Fans (Engine Cooling, Hybrid Battery)
- Pumps (Water, Oil, Fuel)
- Transmission Actuators
- Instrumentation Cluster/ Center Stack
- HVAC Motor Control
- Infotainment
- Body Control

### Features:

- Automotive grade parts - AEC qualified
- Temperature ranges up to  $T_a = 150^{\circ}\text{C}$
- Small profile, thermally efficient packaging

[www.allegromicro.com/promo1064](http://www.allegromicro.com/promo1064)

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